

Borg 2019 trace

public sample
405,894 rows
(usage windows)

Filter

failed = 0,
end > start
313,216 rows

Job model

$b = \lfloor 100u \rfloor + 1$
u: CPU in NCUs
Borg priority

Workload model (seed 42)

500 jobs; arrivals $\div 10^7$, bursts $\times 10^4 \rightarrow$ offered load $\rho = 30$
synthetic deadlines $D = (1 + s)b$, $s \sim U(0.5, 2)$

Simulator (one core)

FCFS, SJF, RR, Priority,
MLQ, MLFQ, EDF, CFS
load sweep $\rho = 0.5\text{--}30$
RR quantum $q = 1\text{--}10^6$

Kernel Baseline (Linux 7.0)

SCHED_FIFO, SCHED_RR,
SCHED_DEADLINE, EEVDF
(2 slices); 3 replicates
1 unit = 100 μ s of CPU

Waiting, turnaround and response time, context switches,
Jain (waiting, slowdown), deadline miss ratio;
simulator vs. kernel by rank agreement (Kendall τ)